AI INTERVIEW COACH

Document 3 of 3

Requirements Gathering

Stakeholder Analysis, User Stories and Use Cases, Functional Requirements, and Non-Functional Requirements.

Item	Detail
Project	AI Interview Coach
Team	Team 147
Course / Track	AI Capstone Project – Hybrid (Deep Learning / NLP + LLMs & RAG)
Document set	Project Documentation Package (3 documents)
Source baseline	16 October 2026 (Version 1.0 of the source documents)
Status	Consolidated draft for academic review

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Appendix A – Source Conflicts and Open Decisions (Requirements)

About This Document

This document is one of three that consolidate the nine Team 147 source documents (Project Proposal and Plan, Task Assignment and Rules, Risk Assessment and Mitigation Plan, KPIs Framework, Feedback and Evaluation Specification, Requirements extract, Software Requirements Specification, Software Design Document, and Stakeholder Analysis / User Stories / Use Cases). The source documents are treated as the source of truth. Status labels used throughout:

Label	Meaning
Confirmed	Stated in the source documents as a decision, scope item or documented fact.
Proposed	Stated in the sources as a proposal, target or recommendation that the team has not yet confirmed.
Recommendation (review)	Added during this consolidation review; not present in any source document.
To Be Confirmed	Required information (for example a calendar date or an owner) is missing or conflicting.
Not Provided in the Source Documents	No evidence exists in the supplied files.

Requirements Conventions

- **Identifiers:** functional requirements are FR-001 to FR-073 and non-functional requirements NFR-001 to NFR-073. The identifiers of the Requirements extract (FR-01…, NFR-P-01…) are preserved in the “Legacy ID” column for NFR and are a zero-padded renumbering for FR (FR-01 → FR-001). The SRS uses a different area-based scheme (FR-AUTH-01…); its reconciliation is listed in Appendix A.
- **Basis:** “Proposal-supported” means the requirement is directly supported by the Project Proposal; “Proposed” means it was added in the Requirements extract as necessary for a usable, secure system.
- **Numeric targets** in the Requirements extract are all stated to be proposed targets. Where another source gives a different value, the conflict is flagged, not resolved (Appendix A).
- **Priority:** High / Medium / Low as in the Requirements extract. The SRS uses MoSCoW (M / S / C); no mapping between the two schemes is defined in the sources.
- Scope note: there is no administrator interface. Administrative monitoring is limited to AWS CloudWatch (FR-071 to FR-073).

3.1 Stakeholder Analysis

Stakeholders are drawn from the Stakeholder Analysis document and the SRS user classes. The SRS defines a single end-user class (the Candidate); there is no administrator role. “Influence” and “Interest” ratings are Recommendation (review) assessments made for this consolidation and are not source data.

Stakeholder	Type	Role and interests	Needs and expectations	Influence / interest (review)
Job seekers, students and recent graduates (Candidate)	Direct – primary user	Practise interviews; beneficiaries of the platform.	Realistic practice; personalized CV-aware feedback; objective self-assessment (confidence score, strengths, weaknesses, suggestions).	Medium / High
Mennatullah Essam – Cloud / DevOps Engineer	Direct – project team; operator	Deploys and monitors the platform; the only operational “user” (CloudWatch).	Reliable AWS deployment (EC2, S3, RDS, API Gateway); monitoring of performance via CloudWatch.	High / High
Sawsan Mahmoud Elgamal – Backend Developer	Direct – project team	Owns the API and integration.	Clear API contract between frontend and AI modules (FastAPI).	High / High
George Ehab Sobhy Dos – Machine Learning Engineer	Direct – project team	Owns speech-to-text and scoring.	Whisper and evaluation models (Sentence Transformers / BERT / DistilBERT); human-rated benchmarks to validate the confidence score.	High / High
Ahmed Hamdy Ahmed – Database Engineer	Direct – project team	Owns the data layer.	Schema for users, sessions and results (PostgreSQL / MySQL on AWS RDS).	High / High
Anas Hany Shaker – Frontend Developer	Direct – project team	Owns the user interface.	React.js interface: CV upload, interview session, results dashboard.	High / High
Belal Salah Hassan – GenAI Integration Lead	Direct – project team	Owns generative AI and RAG.	Personalized question generation and ideal-answer suggestions via OpenAI API / Amazon Bedrock.	High / High
Course lecturer / supervisor	Indirect – reviewer / evaluator	Reviews and grades the project; escalation point for team conflicts.	Feedback and evaluation on documentation, implementation, testing and presentation (see Document 2).	High / Medium
GenAI and cloud providers (OpenAI, Amazon Bedrock, AWS)	Indirect – external service providers	Supply APIs and infrastructure under their own terms.	Use within terms, quotas and budget.	Medium / Low
Sources of interview content and job data	Indirect – external data providers	Public job postings (APIs / scraping) and curated question banks and career resources for RAG.	Use consistent with site terms and licences.	Low / Low

Table 3.1 – Stakeholder matrix (review assessment)

	High interest	Low interest
High influence	Manage closely: project team members, DevOps operator, course lecturer (medium interest)	Keep satisfied: —
Medium / low influence	Keep informed: job seekers (primary users)	Monitor: GenAI / cloud providers, content and data sources

3.2 User Stories and Use Cases

3.2.1 User Stories

Format: As a [role], I want [goal], so that [benefit]. US-01 to US-13 are taken from the Stakeholder Analysis document. US-14 to US-16 were added during consolidation (Recommendation (review)) because the functional requirements FR-001 to FR-006, FR-028 and FR-069 had no corresponding story.

ID	User story	Acceptance criteria	Basis
US-01	As a Job seeker, I want to upload my CV (PDF or DOCX), so that the system can tailor the interview to my background.	CV is accepted in PDF/DOCX and converted to text; user is told if the file cannot be processed.	Source
US-02	As a Job seeker, I want to select my target job role, so that the questions and feedback match the job I am preparing for.	A role can be chosen before a session starts and is used for question generation and evaluation.	Source
US-03	As a Job seeker, I want to receive interview questions personalized to my CV and target role, so that my practice reflects a realistic interview.	Questions are generated from the uploaded CV and selected role (GenAI layer).	Source
US-04	As a Job seeker, I want to type my answer to each interview question, so that I can practice responding.	A text answer can be submitted for each question within a session.	Source
US-05	As a Job seeker, I want to receive a confidence score for each answer, so that I have an objective measure of how well I answered.	A score is returned for every submitted answer, based on semantic similarity to an ideal response.	Source
US-06	As a Job seeker, I want to see the strengths and weaknesses of my answers, so that I know what to keep doing and what to fix.	Strengths and weaknesses are shown alongside the score.	Source
US-07	As a Job seeker, I want to get suggestions on how to improve my answers, so that I can improve in my next attempt.	Actionable improvement suggestions are displayed with the feedback.	Source
US-08	As a Job seeker, I want to see an ideal answer suggestion for a question, so that I can compare it with my own answer.	An ideal answer, generated for my CV and role, can be viewed after I answer.	Source
US-09	As a Job seeker, I want to be asked personalized follow-up questions, so that the session feels like a real interview.	Follow-up questions are generated after an answer is evaluated.	Source
US-10	As a Job seeker, I want to view my results on a dashboard, so that I can review my performance in one place.	Scores, strengths, weaknesses, and suggestions are displayed on a results dashboard.	Source
US-11	As a Job seeker, I want to have my sessions and results saved to my account, so that I can come back to them later.	Users, sessions, and results are stored in the database.	Source; Inferred from the database schema in the Proposal; account / login details are not specified in the Proposal (see US-14).
US-12	As a Job seeker, I want to answer by speaking instead of typing, so that I can practice speaking as in a real interview.	Speech is transcribed with Whisper and then evaluated like a typed answer.	Source; Optional / stretch (Appendix A, C-03).
US-13	As a DevOps engineer, I want to monitor system performance in CloudWatch, so that I can detect and fix problems with the deployed platform.	Performance of the deployed system is visible in AWS CloudWatch.	Source; Operational story; the DevOps engineer is a team member, not an end user.
US-14	As a Job seeker, I want to register and log in with my email and password, so that my CV, sessions and results are private to me.	Registration requires name, email and password; duplicate email is rejected; login grants access to the user's own data only; after logout protected pages are inaccessible.	Added (review)
US-15	As a Job seeker, I want to delete my CV, sessions or account, so that I control my personal data.	Deletion requires explicit confirmation; the CV, sessions, answers and results are removed.	Added (review)

ID	User story	Acceptance criteria	Basis
US-16	As a Job seeker, I want to resume an interrupted session, so that I do not lose my submitted answers.	Resume starts at the first unanswered question; previously submitted answers are present.	Added (review)

3.2.2 Use Case Summary

ID	Use case	Primary actor	User stories	Functional requirements
UC-01	Upload CV and select target role	Job seeker	US-01, US-02	FR-007 to FR-017
UC-02	Practice interview session	Job seeker (with AI evaluation module and GenAI service)	US-03 to US-09, US-12, US-16	FR-018 to FR-060
UC-03	View results dashboard and history	Job seeker	US-10, US-11, US-15	FR-061 to FR-070
UC-04	Monitor system performance	DevOps engineer	US-13	FR-071 to FR-073
UC-05	Register and log in (added in review)	Job seeker (guest)	US-14, US-15	FR-001 to FR-006

3.2.3 Use Case Descriptions

UC-01: Upload CV and select target role

Element	Description
Actor	Job seeker
Preconditions	User has access to the web application and is logged in (FR-007).
Trigger	The user opens the CV upload screen.
Main flow	1. User opens the React.js interface. 2. User uploads a CV (PDF/DOCX). 3. System validates the file, converts the CV to text and stores the data. 4. User selects a target job role (optionally pastes a job description). 5. System confirms and prepares the interview session.
Alternative flows	Unsupported, oversized or unreadable file: the system rejects it, states the accepted formats and size limit, and asks for another file (FR-008, FR-011).
Expected outcome	CV text and target role are stored and available to the session.

UC-02: Practice interview session

Element	Description
Actor	Job seeker (with the AI evaluation module and GenAI service)
Preconditions	UC-01 completed (processed CV and selected role; FR-024).
Trigger	The user starts a session.
Main flow	1. System generates personalized questions from the CV and target role (OpenAI API / Amazon Bedrock). 2. System displays a question. 3. User types an answer and submits it. 4. FastAPI backend receives and stores the answer. 5. Sentence Transformers / BERT / DistilBERT compare the answer with the reference answer and compute a confidence score, strengths and weaknesses. 6. GenAI layer produces improvement suggestions, an ideal answer and a follow-up question. 7. System returns and displays the feedback. 8. Steps 2–7 repeat until all questions are answered or the user ends the session.

Element	Description
Alternative flows	GenAI or evaluation service unavailable: the system informs the user, falls back to the question bank where possible, and the session can be retried (NFR-054 to NFR-056). Voice answer (if supported): audio is transcribed with Whisper before step 5 and the user may edit the transcript (FR-035 to FR-038). User skips a question or ends early: results are generated from the answered questions (FR-027, FR-030).
Expected outcome	Answers, scores and feedback are saved for the session.

UC-03: View results dashboard and history

Element	Description
Actor	Job seeker
Preconditions	At least one answer has been evaluated.
Trigger	The user opens the results dashboard or session history.
Main flow	1. User opens the results dashboard. 2. System retrieves session results from the database. 3. System displays confidence scores, strengths, weaknesses and improvement suggestions. 4. User may open past sessions from the history list.
Alternative flows	No results yet: the system prompts the user to start a practice session.
Expected outcome	User has reviewed their performance.

UC-04: Monitor system performance

Element	Description
Actor	DevOps engineer
Preconditions	System is deployed on AWS.
Trigger	Routine review or a CloudWatch alarm.
Main flow	1. DevOps engineer opens AWS CloudWatch. 2. System metrics and logs are reviewed. 3. Issues are identified and addressed.
Alternative flows	None documented in the sources.
Expected outcome	System health is known to the team.

UC-05: Register and log in (added in review)

Element	Description
Actor	Job seeker (guest)
Preconditions	None.
Trigger	The user opens the application without a session.
Main flow	1. Guest registers with name, email and password. 2. System validates the input and creates the account. 3. User logs in with email and password. 4. System grants access to the user's own data only.
Alternative flows	Email already registered: registration is rejected with a message (FR-002). Wrong password: access denied; repeated failures trigger a temporary block (NFR-018). Forgotten password: reset link (FR-005).
Expected outcome	Authenticated session; protected pages and endpoints are accessible.

3.2.4 Actor–Use Case Matrix

Actor	UC-01	UC-02	UC-03	UC-04	UC-05
Job seeker	●	●	●		●
DevOps engineer				●	
AI evaluation module / GenAI service (supporting)		●			

The matrix is a textual alternative to a use case diagram; it is derived directly from the use cases above.

3.3 Functional Requirements

73 functional requirements are consolidated from the Requirements extract, grouped by module. Each is a verifiable “shall” statement; user-visible acceptance criteria are given in the user stories (3.2.1). No duplicates were found within the Requirements extract; the one-to-one reconciliation with the SRS area-coded requirements is still pending (Appendix A, C-R08). The Trace column links each requirement to user stories (US) and use cases (UC).

ID	Requirement	Description	Priority	Basis	Trace
3.3.1 User Account & Authentication					
FR-001	User Registration	The system shall allow a visitor to create an account by providing a name, email address, and password.	High	Proposed (not in Proposal)	US-14, US-15; UC-05
FR-002	Unique Email Enforcement	The system shall reject registration when the email address is already registered and shall display a message stating so.	High	Proposed (not in Proposal)	US-14, US-15; UC-05
FR-003	User Login	The system shall authenticate a user with a valid email and password and grant access to that user's own data only.	High	Proposed (not in Proposal)	US-14, US-15; UC-05
FR-004	User Logout	The system shall allow a logged-in user to log out, after which protected pages and API endpoints shall be inaccessible without logging in again.	High	Proposed (not in Proposal)	US-14, US-15; UC-05
FR-005	Password Reset	The system shall allow a user who forgot their password to set a new one through a time-limited reset link sent to their registered email.	Low	Proposed (not in Proposal)	US-14, US-15; UC-05
FR-006	Account Deletion	The system shall allow a user to delete their account, which removes the user's CV, sessions, answers, and results (see NFR-027).	Medium	Proposed (not in Proposal)	US-14, US-15; UC-05
3.3.2 CV Upload & Processing					
FR-007	CV Upload	The system shall allow a logged-in user to upload a CV file through the web interface.	High	Proposal-supported	US-01; UC-01
FR-008	CV File Validation	The system shall accept only PDF and DOCX CV files not larger than 5 MB and shall reject other files with a message naming the accepted formats and size limit.	High	Proposal-supported	US-01; UC-01
FR-009	CV Text Extraction	The system shall convert an uploaded CV into plain text for downstream processing.	High	Proposal-supported	US-01; UC-01
FR-010	CV Processing Status	The system shall display the CV processing status (uploading, processing, completed, or failed) to the user.	Medium	Proposed (not in Proposal)	US-01; UC-01
FR-011	CV Processing Failure Notification	The system shall notify the user when no text can be extracted from a CV (for example, a corrupted or image-only file) and shall allow the user to upload a different file.	High	Proposed (not in Proposal)	US-01; UC-01
FR-012	Extracted Text Preview	The system shall allow the user to view the extracted CV text before starting an interview session.	Low	Proposed (not in Proposal)	US-01; UC-01
FR-013	CV Replacement and Deletion	The system shall allow a user to replace or delete a previously uploaded CV.	Medium	Proposed (not in Proposal)	US-01; UC-01
FR-014	CV Reuse	The system shall allow a user to start new sessions using a previously uploaded CV without re-uploading it.	Medium	Proposed (not in Proposal)	US-01; UC-01
3.3.3 Job Role Selection					
FR-015	Job Role List	The system shall present the user with a predefined list of target job roles.	High	Proposal-supported	US-02; UC-01
FR-016	Job Role Selection	The system shall require the user to select exactly one target job role before a session can start.	High	Proposal-supported	US-02; UC-01
FR-017	Optional Job Description Input	The system shall allow the user to optionally paste a job description of up to 5,000 characters to further tailor the interview questions.	Low	Proposal-supported	US-02; UC-01
3.3.4 Interview Question Generation					
FR-018	Personalized Question Generation	The system shall generate interview questions based on the user's CV content and selected target role.	High	Proposal-supported	US-03; UC-02
FR-019	Question Count	The system shall generate between 5 and 10 questions per session, with a default of 5.	Medium	Proposed (not in Proposal)	US-03; UC-02

ID	Requirement	Description	Priority	Basis	Trace
FR-020	Question Categories	The system shall include both technical and behavioral questions in each generated question set when the role supports both.	Medium	Proposed (not in Proposal)	US-03; UC-02
FR-021	Question Bank Grounding	The system shall use the curated interview question bank and career resources as reference material when generating questions.	Medium	Proposal-supported	US-03; UC-02
FR-022	Duplicate Question Prevention	The system shall not present the same question more than once within a session.	Medium	Proposed (not in Proposal)	US-03; UC-02
FR-023	Question Persistence	The system shall store each generated question with its session so the question set remains unchanged if the session is resumed.	High	Proposed (not in Proposal)	US-03; UC-02
3.3.5 Interview Session Management					
FR-024	Session Creation	The system shall create a new interview session only when the user has a processed CV and a selected target role.	High	Proposal-supported	US-04, US-16; UC-02
FR-025	Sequential Question Presentation	The system shall present session questions one at a time in a fixed order.	High	Proposed (not in Proposal)	US-04, US-16; UC-02
FR-026	Progress Indicator	The system shall display the current question number and total number of questions during a session (for example, "Question 3 of 5").	Medium	Proposed (not in Proposal)	US-04, US-16; UC-02
FR-027	Skip Question	The system shall allow the user to skip a question, and skipped questions shall be recorded as unanswered and excluded from the overall score.	Low	Proposed (not in Proposal)	US-04, US-16; UC-02
FR-028	Session Resume	The system shall allow a user to resume an interrupted session from the first unanswered question.	Medium	Proposed (not in Proposal)	US-04, US-16; UC-02
FR-029	Session Completion	The system shall mark a session as completed once all questions are answered or skipped and shall then display the session results.	High	Proposal-supported	US-04, US-16; UC-02
FR-030	End Session Early	The system shall allow a user to end a session before all questions are answered and shall generate results from the answered questions only.	Low	Proposed (not in Proposal)	US-04, US-16; UC-02
3.3.6 Answer Submission					
FR-031	Text Answer Submission	The system shall allow the user to submit a typed text answer to each interview question.	High	Proposal-supported	US-04; UC-02
FR-032	Answer Length Limits	The system shall reject empty answers and answers longer than 2,000 characters and shall tell the user which limit was violated.	Medium	Proposed (not in Proposal)	US-04; UC-02
FR-033	Answer Review Before Submission	The system shall allow the user to edit an answer before submitting it.	Low	Proposed (not in Proposal)	US-04; UC-02
FR-034	Answer Persistence	The system shall store each submitted answer linked to its user, session, and question.	High	Proposal-supported	US-04; UC-02
3.3.7 Speech-to-Text					
FR-035	Spoken Answer Input	The system shall allow the user to provide an answer by recording audio in the browser or uploading an audio file as an alternative to typing. Open decision C-03: voice input is a stretch goal in the Proposal and Task Assignment.	Medium	Proposal-supported	US-12; UC-02
FR-036	Speech Transcription	The system shall transcribe submitted audio answers into text using the Whisper model. Open decision C-03.	Medium	Proposal-supported	US-12; UC-02
FR-037	Transcript Review	The system shall display the transcribed text to the user and allow edits before the answer is submitted for evaluation. Open decision C-03.	Medium	Proposed (not in Proposal)	US-12; UC-02
FR-038	Audio Input Error Handling	The system shall notify the user and offer text input when audio cannot be transcribed or has an unsupported format or duration. Open decision C-03.	Medium	Proposed (not in Proposal)	US-12; UC-02
3.3.8 AI Answer Evaluation					
FR-039	Answer Evaluation	The system shall evaluate each submitted answer against the reference answer for its question using the Sentence Transformers / BERT / DistilBERT evaluation model.	High	Proposal-supported	US-05; UC-02

ID	Requirement	Description	Priority	Basis	Trace
FR-040	Reference Answer Source	The system shall obtain the reference answer for each question from the curated question bank or from the GenAI Service.	High	Proposal-supported	US-05; UC-02
FR-041	Per-Answer Evaluation Result	The system shall return the evaluation result (score, strengths, weaknesses, suggestions) for each answer after it is submitted.	High	Proposal-supported	US-05; UC-02
FR-042	Non-Substantive Answer Handling	The system shall assign a low score to answers that are off-topic or that contain fewer than 10 words, and shall tell the user that the answer was too brief or irrelevant.	Medium	Proposed (not in Proposal)	US-05; UC-02
3.3.9 Confidence / Semantic Similarity Scoring					
FR-043	Semantic Similarity Computation	The system shall compute the semantic similarity between the user's answer and the reference answer using embedding-based cosine similarity.	High	Proposal-supported	US-05; UC-02
FR-044	Confidence Score Generation	The system shall convert the similarity result into a Confidence Score on a 0–100 scale for each answered question.	High	Proposal-supported	US-05; UC-02
FR-045	Score Band Label	The system shall assign each Confidence Score a label (Needs Improvement, Fair, Good, or Excellent) based on defined score ranges.	Medium	Proposed (not in Proposal)	US-05; UC-02
FR-046	Overall Session Score	The system shall compute the overall session score as the average Confidence Score of all answered questions.	High	Proposed (not in Proposal)	US-05; UC-02
FR-047	Scoring Consistency	The system shall produce the same Confidence Score when the same answer is evaluated against the same reference answer more than once.	Medium	Proposed (not in Proposal)	US-05; UC-02
3.3.10 Strengths & Weaknesses Identification					
FR-048	Per-Answer Strengths	The system shall identify and list at least one strength for each evaluated answer that scores 50 or above.	High	Proposal-supported	US-06; UC-02, UC-03
FR-049	Per-Answer Weaknesses	The system shall identify and list at least one weakness for each evaluated answer that scores below 80.	High	Proposal-supported	US-06; UC-02, UC-03
FR-050	Session-Level Strengths and Weaknesses	The system shall summarize the user's main strengths and weaknesses across all answered questions in a session.	High	Proposal-supported	US-06; UC-02, UC-03
3.3.11 Personalized Feedback & Recommendations					
FR-051	Per-Answer Improvement Suggestions	The system shall provide at least one actionable improvement suggestion for each answer that scores below 80.	High	Proposal-supported	US-07; UC-02
FR-052	Session-Level Recommendations	The system shall provide a consolidated list of improvement recommendations at the end of each session.	High	Proposal-supported	US-07; UC-02
FR-053	Feedback Traceability	The system shall display each piece of feedback together with the question and answer it relates to.	Medium	Proposed (not in Proposal)	US-07; UC-02
3.3.12 Ideal Answer Generation					
FR-054	Ideal Answer Generation	The system shall generate an ideal answer for each question based on the user's CV and target role using the GenAI Service.	High	Proposal-supported	US-08; UC-02
FR-055	Ideal Answer Reveal	The system shall display the ideal answer only after the user has submitted or skipped the corresponding question.	Medium	Proposed (not in Proposal)	US-08; UC-02
FR-056	AI-Generated Content Label	The system shall label ideal answers and generated questions as AI-generated.	Low	Proposed (not in Proposal)	US-08; UC-02
3.3.13 Follow-up Question Generation					
FR-057	Follow-up Question Generation	The system shall generate a personalized follow-up question based on the user's answer and CV using the GenAI Service.	Medium	Proposal-supported	US-09; UC-02
FR-058	Follow-up Question Response	The system shall allow the user to answer or decline a follow-up question.	Medium	Proposed (not in Proposal)	US-09; UC-02
FR-059	Follow-up Evaluation	The system shall evaluate and score answers to follow-up questions using the same method as main questions.	Medium	Proposed (not in Proposal)	US-09; UC-02
FR-060	Follow-up Limit	The system shall generate at most one follow-up question per main question.	Low	Proposed (not in Proposal)	US-09; UC-02
3.3.14 Results Dashboard					

ID	Requirement	Description	Priority	Basis	Trace
FR-061	Session Results Summary	The system shall display the overall score, number of questions answered, target role, and session date on the results dashboard.	High	Proposal-supported	US-10; UC-03
FR-062	Per-Question Breakdown	The system shall display, for each question, the user's answer, Confidence Score, strengths, weaknesses, and suggestions.	High	Proposal-supported	US-10; UC-03
FR-063	Score Visualization	The system shall display scores graphically (for example, a score gauge and a per-question bar chart) in addition to numeric values.	Medium	Proposal-supported	US-10; UC-03
FR-064	Strengths and Weaknesses Display	The system shall display session-level strengths, weaknesses, and recommendations in clearly separated sections.	High	Proposal-supported	US-10; UC-03
FR-065	Answer Comparison View	The system shall allow the user to view their answer and the ideal answer side by side for each question.	Medium	Proposed (not in Proposal)	US-10; UC-03
3.3.15 Data Storage & Session History					
FR-066	Results Persistence	The system shall store users, CV text, selected roles, sessions, questions, answers, scores, and feedback in the PostgreSQL/MySQL database on AWS RDS.	High	Proposal-supported	US-11, US-15; UC-03
FR-067	Session History List	The system shall display a list of the user's previous sessions with date, target role, and overall score, sorted from newest to oldest.	Medium	Proposed (not in Proposal)	US-11, US-15; UC-03
FR-068	Past Session Review	The system shall allow the user to open the full results of any previous completed session.	Medium	Proposed (not in Proposal)	US-11, US-15; UC-03
FR-069	Session Deletion	The system shall allow a user to delete a past session and all of its associated answers and results.	Low	Proposed (not in Proposal)	US-11, US-15; UC-03
FR-070	Score Trend Display	The system shall display the user's overall scores across their completed sessions in chronological order.	Low	Proposed (not in Proposal)	US-11, US-15; UC-03
3.3.16 Monitoring					
FR-071	Application Logging	The system shall send backend application logs, including errors and API request events, to AWS CloudWatch.	High	Proposal-supported	US-13; UC-04
FR-072	Performance Metrics	The system shall publish request count, response time, and error rate metrics to AWS CloudWatch.	Medium	Proposal-supported	US-13; UC-04
FR-073	AI Component Failure Logging	The system shall log each failure of the CV processing, speech-to-text, evaluation, and GenAI Service components with a timestamp and error type.	Medium	Proposed (not in Proposal)	US-13; UC-04

3.4 Non-Functional Requirements

71 non-functional requirements are consolidated from the Requirements extract (NFR-001 to NFR-071), and two further maintainability requirements are added from the Task Assignment (NFR-072, NFR-073). **All numeric targets are Proposed targets**: the Proposal gives no performance, security or availability numbers, and no compliance obligation or security guarantee is asserted. Where another source states a different value, the Notes column flags the conflict (Appendix A).

ID	Legacy ID	Requirement	Metric / acceptance criteria (Proposed)	Priority	Notes
3.4.1 Performance and Scalability					
NFR-001	NFR-P-01	The system shall return standard API responses (login, history, session retrieval) quickly under normal load.	95% of requests complete within 2 seconds, excluding AI-processing endpoints.	High	Differs from KPI Framework (p95 ≤ 200 ms) and Task Assignment (p95 ≤ 500 ms); C-R04.
NFR-002	NFR-P-02	The system shall load the main frontend pages quickly.	Initial page load within 3 seconds on a 10 Mbps connection.	Medium	—
NFR-003	NFR-P-03	The system shall process an uploaded CV (text extraction and storage) promptly.	95% of CVs up to 5 MB are processed within 10 seconds.	High	—
NFR-004	NFR-P-04	The system shall generate a session's question set within an acceptable time.	95% of question sets (5–10 questions) are generated within 20 seconds, including the GenAI Service call.	High	—
NFR-005	NFR-P-05	The system shall return the AI evaluation result for a submitted text answer within an acceptable time.	95% of evaluations (embedding, scoring, strengths and weaknesses) are returned within 5 seconds.	High	KPI Framework, Task Assignment and Risk Plan use p95 ≤ 3 s; C-R04.
NFR-006	NFR-P-06	The system shall return generated ideal answers and follow-up questions within an acceptable time.	95% are returned within 10 seconds each.	Medium	—
NFR-007	NFR-P-07	The system shall transcribe spoken answers promptly.	A 60-second audio clip is transcribed within 20 seconds in 95% of cases.	Medium	—
NFR-008	NFR-P-08	The system shall support a realistic number of simultaneous users.	25 concurrent users with all NFR-001 to NFR-006 targets still met.	Medium	Task Assignment uses 20 concurrent sessions; C-R05.
NFR-009	NFR-P-09	The database shall respond quickly to common queries.	95% of standard queries (session lookup, history list) complete within 200 ms with 10,000 stored sessions.	Medium	—
NFR-010	NFR-P-10	The system shall keep resource usage within safe limits during normal operation.	Average CPU and memory utilization on application servers stay below 80% during a 25-user load test.	Medium	—
NFR-011	NFR-P-11	The AI models shall not be reloaded for every request.	Models are loaded once at service start-up, and per-request model loading time is 0 seconds.	Medium	—
NFR-012	NFR-P-12	The system shall allow capacity to be increased without redesign.	Application capacity can be increased by adding or resizing AWS compute resources without code changes.	Low	—
3.4.2 Security and Privacy					
NFR-013	NFR-S-01	The system shall require authentication for all endpoints that access user data.	Requests without a valid token to protected endpoints return an authentication error (HTTP 401).	High	—
NFR-014	NFR-S-02	The system shall enforce per-user authorization so that users can access only their own CVs, sessions, and results.	A user requesting another user's resource ID receives HTTP 403 or 404 in 100% of tested cases.	High	—
NFR-015	NFR-S-03	The system shall store passwords only as salted hashes produced by a one-way adaptive hashing algorithm.	No plaintext or reversibly encrypted password is present in the database or logs.	High	—
NFR-016	NFR-S-04	The system shall enforce a minimum password policy.	Passwords shorter than 8 characters or without both letters and digits are rejected.	Medium	SRS FR-AUTH-02 requires ≥ 8 characters with upper-case, lower-case and digit; C-R02.

ID	Legacy ID	Requirement	Metric / acceptance criteria (Proposed)	Priority	Notes
NFR-017	NFR-S-05	The system shall expire authenticated sessions.	Access tokens expire within 60 minutes. A logged-out token is rejected afterwards.	High	SRS FR-AUTH-03 uses a 15-minute access token and 7-day refresh token; C-R01.
NFR-018	NFR-S-06	The system shall limit repeated failed login attempts.	After 5 failed attempts within 10 minutes, further attempts for that account are blocked for 15 minutes.	Medium	SRS FR-AUTH-06 uses 5 failures in 15 minutes with a 15-minute block; C-R03.
NFR-019	NFR-S-07	The system shall encrypt all communication between clients and the system.	All endpoints are reachable only over HTTPS (TLS 1.2 or higher). HTTP requests are redirected or refused.	High	—
NFR-020	NFR-S-08	The system shall encrypt stored personal data at rest.	Encryption at rest is enabled on the AWS RDS instance and on any S3 storage holding CVs.	High	—
NFR-021	NFR-S-09	The system shall protect stored CV files from public access.	CV files are not accessible by public URL. An unauthenticated request to a CV location is denied.	High	—
NFR-022	NFR-S-10	The database shall not be directly accessible from the public internet.	The database accepts connections only from the backend's network and rejects connections from other sources.	High	—
NFR-023	NFR-S-11	The system shall validate all user input on the server side.	Invalid, oversized, or malformed inputs (fields, files, answers) are rejected with a validation error and are never processed.	High	—
NFR-024	NFR-S-12	The system shall verify uploaded file content, not only file extension.	A non-PDF/DOCX file renamed with a .pdf or .docx extension is rejected.	Medium	—
NFR-025	NFR-S-13	The system shall protect against SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF) where applicable.	Testing with standard injection and script payloads shows no executed injected code or altered queries.	High	—
NFR-026	NFR-S-14	The system shall limit API request rates per user or IP address.	Requests above a configured threshold (for example, 100 per minute) receive HTTP 429.	Medium	—
NFR-027	NFR-S-15	The system shall permanently delete a user's data when the user deletes their account or a session.	After deletion, the data cannot be retrieved through the application. Deletion in the primary database completes within 24 hours.	High	—
NFR-028	NFR-S-16	The system shall not retain audio recordings after transcription.	Audio files are discarded once transcription completes or fails.	Medium	—
NFR-029	NFR-S-17	The system shall keep API keys, database credentials, and other secrets out of source code and the frontend.	A scan of the repository and the frontend bundle finds no secrets.	High	—
NFR-030	NFR-S-18	The system shall call the OpenAI API / Amazon Bedrock only from the backend.	No GenAI Service request originates from the browser.	High	—
NFR-031	NFR-S-19	The system shall remove direct contact details (email addresses and phone numbers) from CV content before sending it to the GenAI Service.	Test CVs containing these details produce GenAI requests with those details removed.	Medium	—
NFR-032	NFR-S-20	The system shall exclude passwords, tokens, and CV or answer text from application logs.	A review of CloudWatch logs from a test run finds none of these items.	Medium	—
NFR-033	NFR-S-21	The system shall restrict cross-origin requests to the approved frontend origin.	Requests from other origins are rejected.	Medium	—
NFR-034	NFR-S-22	The system shall not expose technical details (stack traces, database errors) in user-facing error responses.	Forced error conditions return generic error messages only.	Medium	—
NFR-035	NFR-S-23	The system shall grant AWS service permissions on a least-privilege basis.	Each AWS component's role includes only the permissions it needs, as confirmed by an access review.	Medium	—
NFR-036	NFR-S-24	The system shall inform users that their CV and answers are processed by a third-party GenAI Service.	A notice is shown before the first CV upload, and the user must acknowledge it.	Medium	—
3.4.3 Usability and Accessibility					

ID	Legacy ID	Requirement	Metric / acceptance criteria (Proposed)	Priority	Notes
NFR-037	NFR-U-01	The system shall provide a clear, visible navigation structure.	Core tasks (upload CV, start session, view history) are reachable from the main dashboard within 3 clicks.	High	—
NFR-038	NFR-U-02	The system shall show the user where they are in the workflow.	A step indicator (CV → Role → Interview → Results) is visible on every workflow page.	Medium	—
NFR-039	NFR-U-03	The CV upload page shall state the accepted formats and size limit before upload.	Accepted formats and the 5 MB limit are visible on the upload page without user interaction.	High	—
NFR-040	NFR-U-04	The system shall give clear feedback during CV upload and processing.	A progress or status indicator is shown throughout, with an explicit success or failure message at the end.	High	—
NFR-041	NFR-U-05	The system shall show a loading indicator for any operation lasting more than 2 seconds.	Loading indicators appear for question generation, evaluation, and transcription in 100% of cases.	High	—
NFR-042	NFR-U-06	The interview session shall show the current question, progress, and available actions without scrolling on common screens.	Question text, answer input, and submit button are visible without scrolling at 1366×768 resolution.	Medium	—
NFR-043	NFR-U-07	The system shall present feedback in a consistent, labeled structure.	For every answer, the Score, Strengths, Weaknesses, and Suggestions each appear under separate, labeled headings.	High	—
NFR-044	NFR-U-08	The system shall not rely on color alone to communicate scores.	Each score is shown as a number and a text label (for example, "Good") in addition to any color.	Medium	—
NFR-045	NFR-U-09	Charts on the results dashboard shall include labeled axes or values.	Every chart shows score values and category labels.	Medium	—
NFR-046	NFR-U-10	Error messages shall be written in plain language and state how to fix the problem.	Every user-facing error message includes what went wrong and a suggested next step.	High	—
NFR-047	NFR-U-11	The user interface shall be responsive.	All pages are usable without horizontal scrolling at widths from 360 px to 1920 px in the latest versions of Chrome, Edge, Firefox, and Safari.	Medium	—
NFR-048	NFR-U-12	The interface shall meet basic accessibility measures.	All form fields have labels, all functions are operable by keyboard, and normal text has a contrast ratio of at least 4.5:1.	Medium	—
NFR-049	NFR-U-13	The interface shall use consistent layout, terminology, and controls.	Shared UI components (buttons, forms, cards) are reused across pages, and the same terms (for example, "Confidence Score") are used everywhere.	Medium	—
NFR-050	NFR-U-14	New users shall be able to complete their first session without external instructions.	In a test with at least 5 first-time users, at least 80% complete CV upload, role selection, and one answered question without assistance.	Medium	—
NFR-051	NFR-U-15	The system shall ask for confirmation before actions that discard data.	Deleting a CV, session, or account requires an explicit confirmation step.	Medium	—
NFR-052	NFR-U-16	The system shall explain a denied microphone permission.	If microphone access is denied, a message explains how to enable it and offers text input.	Low	—
3.4.4 Reliability, Availability and Recoverability					
NFR-053	NFR-R-01	The system shall be available during the project's demonstration and evaluation period.	At least 95% availability, measured over the evaluation period, excluding announced maintenance.	High	KPI Framework states ≥ 99.5% monthly availability; C-R06.
NFR-054	NFR-R-02	The system shall handle an unavailable or slow GenAI Service without crashing.	GenAI requests time out after 30 seconds and are retried up to 2 times. If still failing, the user sees a clear message and the system continues operating.	High	—
NFR-055	NFR-R-03	The system shall fall back to curated question bank content when question generation via the GenAI Service fails.	During a simulated GenAI outage, a session can still start using questions drawn from the question bank.	High	—
NFR-056	NFR-R-04	Answer evaluation shall continue to work when the GenAI Service is unavailable.	During a simulated outage, submitted answers are still scored against question-bank reference answers where available. Where none is available, the user is told an ideal answer cannot be shown.	Medium	—

ID	Legacy ID	Requirement	Metric / acceptance criteria (Proposed)	Priority	Notes
NFR-057	NFR-R-05	The system shall handle CV processing failures without leaving partial data.	A failed CV processing attempt creates no usable CV record, shows an error, and allows re-upload.	High	—
NFR-058	NFR-R-06	The system shall keep a submitted answer even if its evaluation fails.	After a simulated evaluation failure, the answer is stored, the user is notified, and re-evaluation can be requested.	High	—
NFR-059	NFR-R-07	The system shall let a user resume an interrupted session without losing submitted answers.	After a browser close or network drop, 100% of previously submitted answers are present when the session is resumed.	High	—
NFR-060	NFR-R-08	The system shall retain unfinished sessions for a limited time.	Unfinished sessions can be resumed for at least 7 days.	Low	—
NFR-061	NFR-R-09	The system shall store each answer and its evaluation result as a single consistent unit.	A failure during saving leaves no answer without its result, or result without its answer.	High	—
NFR-062	NFR-R-10	The system shall prevent duplicate records from repeated submissions.	Double-clicking Submit stores only one answer for that question.	Medium	—
NFR-063	NFR-R-11	The database shall enforce referential integrity between users, sessions, questions, answers, and results.	Records referencing non-existent parents cannot be created, and database constraints are verified by tests.	High	—
NFR-064	NFR-R-12	Stored overall scores shall match their underlying per-question scores.	Recalculating the average from stored per-question scores yields the stored overall score in 100% of test sessions.	Medium	—
NFR-065	NFR-R-13	The database shall be backed up automatically.	Automated daily AWS RDS backups are enabled with at least 7 days' retention.	High	—
NFR-066	NFR-R-14	The system shall be recoverable from a database backup.	One successful test restore from a backup is performed and documented before final delivery.	Medium	—
NFR-067	NFR-R-15	The system shall fall back to text input if speech-to-text fails.	After a simulated transcription failure, the user can type the answer and continue the session.	Medium	—
NFR-068	NFR-R-16	The backend shall return structured error responses for failures instead of terminating.	Fault-injection tests (invalid input, unavailable component) return error responses, and the service stays running.	High	—
NFR-069	NFR-R-17	The backend service shall recover automatically after a crash.	After a forced process failure, the service is available again within 5 minutes without manual action.	Medium	—
NFR-070	NFR-R-18	The system shall record operational logs and metrics in AWS CloudWatch.	Application errors, request metrics, and AI-component failures appear in CloudWatch within 5 minutes. Logs are retained for at least 30 days.	High	—
NFR-071	NFR-R-19	The system shall alert the team about significant problems.	A CloudWatch alarm notifies the team when the 5xx error rate exceeds 5% over 5 minutes or CPU utilization exceeds 85% for 10 minutes.	Medium	Risk Plan alarm catalogue uses 5xx > 2% (Critical) and CPU > 80%; C-R07.
3.4.5 Maintainability (added from Task Assignment)					
NFR-072	—	Backend and ML utility code shall be covered by automated tests.	Line coverage ≥ 80% on /backend (Task Assignment §2.3, §3.3; KPI Framework §3.1).	Medium	Source: Task Assignment, KPI Framework. Proposed target; interim target ≥ 60% at end of Sprint 2.
NFR-073	—	Code changes shall pass automated quality gates before merge.	CI checks style / lint, tests and secret scan; peer-reviewed pull request (Task Assignment §3.3).	Medium	Source: Task Assignment. Proposed.

> *The SRS also has sections for scalability, maintainability, portability, privacy and accessibility (§4.4–§4.9). They are covered here by NFR-012 (scalability), NFR-072 and NFR-073 (maintainability), NFR-013 to NFR-036 (security and privacy), NFR-047 and NFR-048 (responsive and accessible interface). A line-by-line SRS reconciliation is a pending action (Appendix A, C-R08). Explicit requirements versus recommendations: NFR items derived from the Proposal are explicit; all thresholds are recommendations until confirmed.*

Appendix A – Source Conflicts and Open Decisions (Requirements)

ID	Topic	Conflicting statements	Status
C-R01	Access token lifetime	Requirements extract NFR-S-05 (NFR-017): 60 minutes. SRS FR-AUTH-03: 15-minute access token with 7-day refresh token.	Unresolved; SRS is the more detailed source
C-R02	Password policy	Requirements extract: ≥ 8 characters with letters and digits. SRS FR-AUTH-02: ≥ 8 characters with upper-case, lower-case and digit; bcrypt cost 12.	Unresolved
C-R03	Login lockout	Requirements extract: 5 failures in 10 minutes, blocked 15 minutes. SRS FR-AUTH-06: 5 failures in 15 minutes, blocked 15 minutes.	Unresolved
C-R04	Latency targets	Evaluation: 95% within 5 s (Requirements) versus p95 ≤ 3 s (Task Assignment, KPI Framework, Risk alarms). Lightweight API: 2 s versus 500 ms versus 200 ms.	Unresolved
C-R05	Concurrency	25 concurrent users (Requirements) versus 20 concurrent sessions (Task Assignment).	Unresolved
C-R06	Availability	≥ 95% during evaluation period (Requirements) versus ≥ 99.5% monthly (KPI Framework).	Unresolved
C-R07	Alert thresholds	NFR-R-19: 5xx > 5% over 5 min, CPU > 85% over 10 min. Risk Plan: 5xx > 2% Critical, CPU > 80% over 10 min.	Unresolved
C-R08	Requirement identifiers	Requirements extract: FR-01…FR-73, NFR-P/S/U/R-nn. SRS: FR-<area>-nn, NFR-<area>-nn with MoSCoW priorities. A one-to-one mapping of SRS requirements to FR-001…FR-073 was not completed in this consolidation.	Pending reconciliation
C-R11	Voice input (C-03 in Document 1)	Stretch goal (Proposal, Task Assignment) versus optional in-scope feature (SRS, Requirements). FR-035 to FR-038 retain Medium priority.	Unresolved
C-R09	GenAI optional or core	Proposal labels GenAI optional; the Requirements extract treats generation as core (High) with fallback requirements NFR-055 and NFR-056.	Resolved by Requirements extract; to be confirmed
C-R10	Database	PostgreSQL / MySQL (Requirements, SDD) versus PostgreSQL with pgvector (SRS). Requirements work with either.	Open (Document 1, C-04)